

# The Uncapacitated Facility Location and $k$ -Median Problems

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## The Uncapacitated Facility Location Problem

The **Uncapacitated Facility Location Problem** (UFLP) is as follows:

Given a

- Finite client set  $\mathcal{D}$ ,
- Finite possible facility set  $\mathcal{F}$ ,
- Function  $f : \mathcal{F} \rightarrow \mathbb{R}_+$  where  $f_i$  is the cost of opening facility  $i \in \mathcal{F}$ , and
- Function  $c : \mathcal{F} \times \mathcal{D} \rightarrow \mathbb{R}_+$ , where  $c_{i,j}$  is the service cost of assigning client  $j \in \mathcal{D}$  to facility  $i \in \mathcal{F}$

We wish to find

- a subset  $X \subset \mathcal{F}$  of facilities to **open**, and
- an assignment  $\sigma : \mathcal{D} \rightarrow X$  of clients to open facilities

such that the sum of

- $c_F(X)$ , facility opening costs  $f_i$  of all  $i \in X$ , and
- $c_S(X)$ , The service costs  $c_{\sigma(j),j}$  of all  $j \in \mathcal{D}$  for the given solution

is minimal.

### Metric vs. Non-Metric UFLP

- In many applications, the service costs  $c$  form a **metric** on  $\mathcal{F} \cup \mathcal{D}$ .
- Both the Metric and Non-Metric UFLP are strongly NP-Hard by polynomial reduction to set cover.
- Approximation algorithms for Metric UFLP rely on the **triangle inequality** for their approximation guarantees. In fact, there is no constant-factor approximation to the Non-Metric UFLP unless  $P = NP$ .
- The performance guarantee of a constant-factor approximation to Metric UFLP must be greater than 1.463 unless  $P = NP$  (Guha & Khuller 1999).

## The Dual Fitting Algorithm

- The Dual Fitting Algorithm is a primal-dual algorithm with a worst-case 2-factor performance guarantee due to Jain et al. (2003).
- **In the dual program, decision variables represent client budgets**, and clients must pay service fees and facility opening costs.
- Service costs  $c_{i,j}$  become a fee that client  $j$  has to pay to be assigned to facility  $i$  (if it opens)
- Facility opening costs  $f_i$  become a fee that all clients who have paid their service costs for facility  $i$  must pay, collectively, in order to open the facility.
- For each client, the budget is independently applied to each service cost.
- If a service cost is fully paid, the remainder of the budget is applied to the the opening cost of the corresponding facility.
- **We iteratively increase the budget** of each client in the network **until all clients are assigned**.
- The amount the budget increases between each iteration is given by the minimum value for which a client will be assigned. This helps limit the number of iterations, leading to an efficient running time of  $O(|\mathcal{F}|^2|\mathcal{D}|)$ .
- After each budget increase, we check if any unassigned clients can now afford a facility that is open, and if so, assign the client to that facility.
- When a facility's opening cost is fully met by payments from the aggregate clients, it is opened; we check this each iteration (but it is not necessarily guaranteed each time).
- Any clients – unassigned or assigned – who have contributed to the opening cost are reassigned to the new facility.
- A **factor-revealing LP** can be formulated for the Dual Fitting Algorithm that allows us to more tightly bound the performance of the algorithm.
- The algorithm itself can be further improved.

## Integer Programming Formulation

Let  $y_i$  be a binary variable representing if facility  $i$  is opened for all  $i \in \mathcal{F}$ , and  $x_{i,j}$  be a binary variable representing if client  $j$  is assigned to facility  $i$ , for all  $i \in \mathcal{F}, j \in \mathcal{D}$ . Then the UFLP can be written as the integer program

$$\begin{aligned} \min \quad & \sum_{i \in \mathcal{F}} f_i y_i + \sum_{i \in \mathcal{F}} \sum_{j \in \mathcal{D}} c_{i,j} x_{i,j} \\ \text{s.t.} \quad & x_{i,j} \leq y_i & \forall j \in \mathcal{D}, i \in \mathcal{F} \\ & \sum_{i \in \mathcal{F}} x_{i,j} = 1 & \forall j \in \mathcal{D} \\ & x_{i,j} \in \{0, 1\} & \forall i \in \mathcal{F}, j \in \mathcal{D} \\ & y_i \in \{0, 1\} & \forall i \in \mathcal{F} \end{aligned}$$

**LP-rounding** and **primal-dual** algorithms take advantage of aspects of linear programming, like duality, to obtain good approximations. The first constant-factor (4-factor) approximation for UFLP (Shmoys et al. 1997) was an LP-rounding algorithm.

## Improving Dual Fitting

- **Greedy augmentation** of an approximation to UFLP is the process of iteratively opening a new facility with the maximum ratio between the improvement in service cost and the cost of opening that facility, until there are no facilities with such a ratio strictly greater than 1.
- **Scaling** an instance of UFLP is simply multiplying the facility costs by some fixed positive constant to obtain a modified problem instance.
- Scaling the UFLP by a factor of 1.504, applying the Dual Fitting algorithm, and then performing greedy augmentation gives a solution with a performance guarantee of 1.52 (Mahdian, Ye, & Zhang 2006).
- Subsequently, Dual Fitting has been improved to a 1.5-approximation (Byrka & Aardal 2007), and to a 1.488-approximation (Li 2013).

## The $k$ -Median Problem

- In practice, we often want to limit the total number of open facilities to be less than some fixed constant  $K$ .
- This is known as the **Metric  $k$ -Facility Location Problem**, and can be described by adding the constraint

$$\sum_{i \in \mathcal{F}} y_i \leq K$$

To the UFLP IP formulation.

- In the special case where all opening costs are zero, we have the **Metric  $k$ -Median Problem**.
- Given integral data, a  $\gamma$ -factor approximation to the UFLP gives us a  $2\gamma$ -factor approximation to the  $k$ -median instance of the problem (Jain & Vazirani 2003).
- **Local search methods** can provide approximation guarantees for facility location problems. This is unusual for local search, which has mainly practical applications. This is a strong technique for general metric  $k$ -median problems, giving a 2.375-factor approximation (Charikar & Guha 2005).

## Other Applications

- In the UFLP, there is no limit on the number of clients that can use a facility – each facility is uncapacitated. A related problem, the **Capacitated Facility Location Problem**, models this case.
- In the case of the **Non-Metric  $k$ -Median Problem**, many modern approaches to  $\alpha$ -size-approximation exist that provide reasonably good polynomial-time approximations even though no constant-factor approximations are available (Young 2025).